

Multi-Source Unsupervised Graph Domain Adaptation via Concise Propagation–Transformation Pipeline

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Abstract

Unsupervised graph domain adaptation (UGDA) aims to transfer knowledge from a labeled source graph to an unlabeled target graph, addressing the performance degradation caused by distributional shifts in node attributes and graph structures across domains. Despite recent progress, existing UGDA approaches still face two key challenges: (C1) **Data-level**: Most methods rely on a single source domain, overlooking the complementary knowledge that could be leveraged from multiple sources. (C2) **Model-level**: Many UGDA models emphasize complex, handcrafted Graph neural network (GNN) architectures, while simpler yet effective designs with propagation (**P**) & transformation (**T**) pipeline remain underexplored. To address these challenges, in this paper, we propose a novel approach, which leverages **Concise Propagation–Transformation** pipeline for multi-source unsupervised **Graph Domain Adaptation**, dubbed as **CPT-GDA**, to better capture complementary knowledge from multiple sources in an efficient manner. Specifically, the proposed CPT-GDA adopts a dual-branch GNN architecture with different depths of propagation but the same P-T patterns, which enables the model to efficiently learn node representations to mitigate domain discrepancy. Meanwhile, to facilitate effective knowledge transfer across graphs, we derive three optimization objectives: (1) *the classifier loss* to learn discriminative representations; (2) *the alignment loss* weighted by the graph Wasserstein distance to align the structure and feature distribution; and (3) *the pseudo-label loss* to refine target node representations. Extensive experiments on real-world datasets confirm that the proposed method outperforms recent state-of-the-art baselines, demonstrating its effectiveness.

CCS Concepts

• **Computing methodologies** → **Machine learning**.

Keywords

Unsupervised learning, Multi-source graph domain adaptation, Graph neural networks

ACM Reference Format:

Jiayi Wang, Yi Li, Xin Zheng, Junyang Chen, Yanqing Guo^{*}, Alan Wee-Chung Liew, and Shirui Pan. 2026. Multi-Source Unsupervised Graph Domain Adaptation via Concise Propagation–Transformation Pipeline. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3774904.3792079>

1 Introduction

Graph neural networks (GNNs) [12, 15, 43, 45, 55, 56, 59] have emerged as powerful tools for graph representation learning, powering various real-world application scenarios with graph structural data in Web, such as social networks [2, 44, 50], recommender systems [8, 13, 37, 42], and biomolecular analysis [17, 18, 48]. A fundamental problem that affects the robustness and generalization of GNNs lies in the distribution shifts between training and test graph data: a GNN trained on a source domain often fails to perform well on a target domain, particularly when both node attributes and graph structures differ. Hence, we focus on the scientific challenges associated with **Web-related graphs**, which play a crucial role in establishing connections between massive computer-understandable knowledge bases and graph structural data for the World Wide Web-related graph learning. By utilizing GNN models to enhance the various web applications, we aim to learn and tackle the critical challenges of the Web by building a creative and effective Web graph learning framework, enabling the blossoming of the Web as a fundamental infrastructure for real-world applications.

To deal with the domain distribution shift problem, graph domain adaptation (GDA) has received extensive attention, aiming

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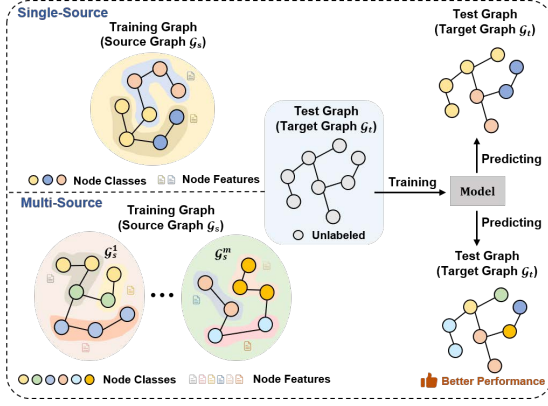


Figure 1: Comparison between single-source unsupervised GDA vs. multi-source unsupervised GDA.

to transfer knowledge from source graph domains to target graph domains, thereby bridging distribution shift discrepancies in both node attributes and graph structures [3, 29, 35, 41, 51, 54, 57, 58]. Depending on the availability of labels in the target domain, existing methods can be broadly divided into supervised GDA [4] and unsupervised GDA [30]. In this study, we focus on unsupervised GDA, as unlabeled target domains are not only more common in practice but also more difficult to handle effectively. Most existing unsupervised GDA methods focus on the single-source adaptation problem, where only one source domain and one target domain are available for training and testing [3, 22, 29, 36, 40]. Typically, by minimizing the distance between source and target graph distributions in the latent feature space, e.g., maximum mean discrepancy (MMD), existing methods aim to learn domain-invariant representations for effective knowledge transferring across graphs with structural and attribute shifts [43, 49].

Despite laying important groundwork, single-source unsupervised GDA faces a critical **data-level challenge (C1): limited complementary knowledge across multiple sources** can be leveraged, which constrains domain adaptation performance. In contrast, multi-source unsupervised GDA can integrate diverse information from heterogeneous sources, enabling the capture of richer domain-invariant representations while mitigating source-specific biases, thereby improving the robustness and generalization ability of GNN models. The difference between multi-source unsupervised GDA and single-source unsupervised GDA is illustrated in Figure 1. However, only limited studies have explored the direction of multi-source unsupervised GDA [51, 54]. Existing approaches either rely on complex meta-learning frameworks with multi-level GNN optimization [54], or fine-tune multiple pretrained GNNs across multi-source domains, both of which incur substantial computational complexity during adaptation. This raises a severe **model-level challenge (C2): complex GNN architectures and optimization schemes** are emphasized to ensure powerful model transferability. In light of this, a natural question arises: *To what extent can mainstream GNNs, with typical message-passing operations, demonstrate inherent robustness and generalization for multi-source unsupervised graph domain adaptation?*

To answer this question and solve the data-level and model-level challenges, in this work, we revisit the importance of typical message-passing operations of GNNs, i.e., disentangled Propagation (P) & Transformation (T). Importantly, we propose a novel approach, which leverages Concise Propagation-Transformation pipeline for multi-source unsupervised Graph Domain Adaptation, dubbed as **CPT-GDA**, to better capture complementary knowledge from multiple sources in an efficient manner. Specifically, the proposed CPT-GDA considers the essential two-fold aspects in P & T pipeline, i.e., depth and pattern, and we first build a dual-branch GNN architecture, where the multi-source branch and the target branch are with different depths of propagation but same P-T patterns. This strategy ensures that the model can capture diverse structural and feature information across domains, so that node representations can be effectively learned to reduce domain divergence. We provide a theoretical analysis to prove the rationality and effectiveness of this dual-branch design. Meanwhile, to effectively optimize the dual-brach GNN, we propose three optimization objectives to guarantee expressive domain adaptation learning, i.e., (1) *the classifier loss* using the source labeled data to learn discriminative representations; (2) *the alignment loss* weighted by the graph Wasserstein distance to align the structure and feature distribution of source and target graphs; and (3) *the pseudo-label loss* between generated target pseudo-labels and predicted class distributions to optimize parameters. Extensive experiments conducted on real-world datasets with cross-graph domains demonstrate the superiority of our proposed framework.

In summary, our main contributions are as follows:

- We propose a multi-source unsupervised graph domain adaptation framework through the concise propagation-transformation pipeline, **CPT-GDA**, to effectively leverage complementary knowledge from multiple source domains, enhancing transferability across diverse graph distributions.
- We build the dual-brach GNN with different depths of propagation but the same fixed P-T patterns, thereby improving its generalization ability for multi-source UGDA.
- We derive three optimization objectives in CPT-GDA: classifier loss, alignment loss, and pseudo label loss, to efficiently explore useful knowledge among multi-source graphs.
- Extensive experimental results on real-world datasets show that CPT-GDA has achieved state-of-the-art performance, further verifying the effectiveness of our method.

2 Related Work

Propagation & Transformation in GNNs. Recently, a series of research efforts [9, 10, 13, 21, 38] have revealed the fact that the entanglement of the Propagation (P) operation and the Transformation (T) operation may affect performance across a wide range of benchmark tasks. These studies also put forward a convincing argument that the true potential of GNNs mainly lies in the P operation rather than the T operation in graph convolution. This understanding stimulates the development of entangled GNNs, intending to separate P and T operations to enhance the overall performance of the model. Following the pioneering work of SGC [38], methods [9, 13] perform P operations in the initial stage and then input the

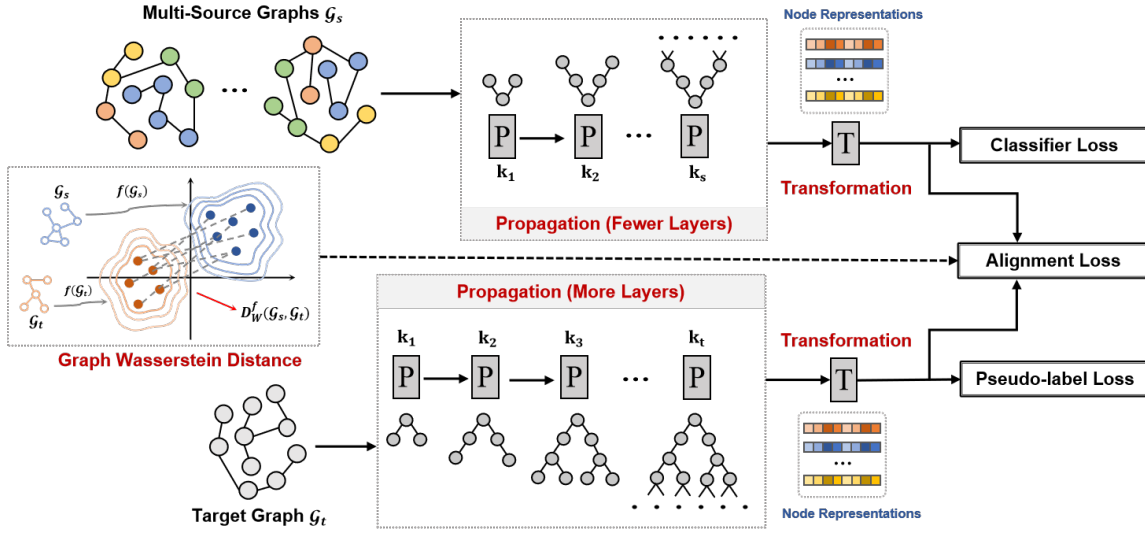


Figure 2: The overall framework of the proposed CPT-GDA.

propagated features into multiple T operations. In contrast, methods such as APPNP [10] and DAGNN [21] first transform the node features using T operations, and then propagate the node information to the remote neighbors using P operations. In this way, the transformed features will be unfolded in the graph, enabling nodes to merge more information from the remote parts in the graph structure.

Unsupervised Graph Domain Adaptation. Existing UGDA methods on graphs can be broadly classified into two categories. Some methods focus on aligning feature distributions between source and target domains based on MMD [24, 40] to better generalize from the source domain to the target domain. Other approaches leverage the graph structure information [23] to guide bootstrapping of neighbors used in GNN computation on the source graphs, which eventually reduces the conditional shift of neighborhoods. In multi-source UGDA, the situation becomes more complex as multiple source domains with different characteristics need to be considered. Several notable works have made significant contributions. Instead of aggregating predictions at the model level like previous multi-source domain adaptation methods, GraphATA [51] aggregates knowledge from multiple pre-trained source models at the node level, dynamically adapting to the local context of each target node to handle complex structural shifts. To learn invariant features for adaptation, SelMAG [54] aligns the target domain to selected source data by minimizing the optimal transport distance and distilling the label function at the classification level.

3 Preliminary

Notations. Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{Y})$, where $\mathcal{V}$ denotes the set of nodes and $\mathcal{E}$ denotes the set of edges. Nodes are associated with a node attribute matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, where n is the number of nodes and d is the dimension of node attributes. The edge structure is characterized by an adjacency matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, where $\mathbf{A}_{i,j} = 1$ indicates the presence of an edge $\mathbf{e}_{i,j} \in \mathcal{E}$ connecting node v_i and

$v_j \in \mathcal{V}$; Otherwise, $\mathbf{A}_{i,j} = 0$. The class information of nodes in $\mathcal{G}$ is represented by $\mathbf{Y} \in \mathbb{R}^{n \times C}$, with C denoting the number of classes. **GNNs with P-T Operations.** Taking the vanilla GCN [15] as a representative example, given a certain layer l , a message passing layer within this framework can be represented in a structured manner as follows:

$$\begin{aligned} \mathbf{P} : \quad \mathbf{M}^l &= \hat{\mathbf{A}} \mathbf{H}^{l-1}, \\ \mathbf{T} : \quad \mathbf{H}^l &= \delta(\mathbf{M}^l \mathbf{W}^l), \end{aligned} \quad (1)$$

where $\hat{\mathbf{A}}$ denotes the normalized adjacent matrix, $\mathbf{H}^{l-1}$ denotes the representation matrix from the previous layer $l-1$ and $\mathbf{H}^0 = \mathbf{X}$. $\mathbf{W}^l$ represents the trainable parameters specific to the l -th layer, $\mathbf{M}^l$ aggregates the information from neighboring nodes formed by message vectors $\mathbf{m}_v^l$ for each node v , and the function $\delta(\cdot)$ is an activation function. By stacking k layers, each node in GCN is capable of utilizing the information from its k -hop neighborhood. Therefore, the model performance is expected to be improved when more distant neighborhood nodes get involved in the training process, as this enables the model to capture more comprehensive and global structural information within the graph.

Problem Definition. Multi-source UGDA aims to transfer knowledge from multiple source GNN models to achieve accurate predictions in the target domain under graph distribution shifts. This involves leveraging labeled knowledge from source domains to adapt to an unlabeled target domain, mitigating inter-domain distribution discrepancies to ensure the GNN generalization performance on the target graph. Specifically, our task can be formalized as:

DEFINITION. [MULTI-SOURCE UNSUPERVISED GRAPH DOMAIN ADAPTATION]. Given M labeled graphs $\mathcal{G}_s = (\mathcal{V}_s^m, \mathcal{E}_s^m, \mathbf{Y}_s^m)_{m=1}^M$ from different source domains and an unlabeled target graph $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t)$, we aim to train a GNN parameterized by θ through $f_\theta : \mathcal{G}_t \rightarrow \mathbf{Y}_t$ using the labeled source graphs and the unlabeled target graph to accurately predict the labels $\mathbf{Y}_t$ for nodes in the target graph $\mathcal{G}_t$.

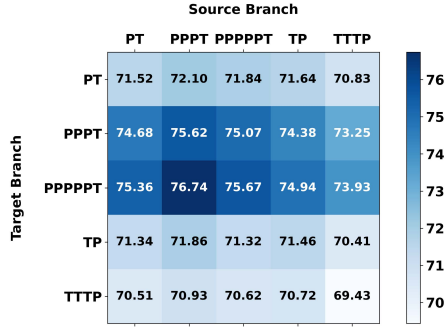


Figure 3: The node classification accuracies (%) of different operations of the source branch and the target branch on the task $C, D \rightarrow A$.

4 Methodology

The overall framework of our proposed CPT-GDA is presented in Figure 2. Given the multi-source graph set $\mathcal{G}_s = (\mathcal{V}_s^m, \mathcal{E}_s^m, \mathbf{Y}_s^m)_{m=1}^M$ and the target graph $\mathcal{G}_t$ as the input, we first build a dual branch GNN architecture formed by the source branch and the target branch, where they follow the same P-T patterns but different depths, i.e., the target branch has more propagation layers. Meanwhile, we calculate the domain-aware similarity between each source graph and the target graph through a graph Wasserstein distance, which is taken as the importance weight to align structural and feature distribution. Then, multi-source graphs are sent to the source branch in sequence, and the target graph is fed into the target branch, learning representative node embeddings with domain invariance. Finally, we minimize the classifier loss, the alignment loss, and the pseudo-label loss by generating target pseudo-labels. More details of each submodule are presented as follows.

4.1 Empirical and Theoretical Analysis for Concise P-T Pipeline

We first analyze how the propagation-transformation pipeline affects the multi-source unsupervised domain adaptation performance on graphs. Our investigations aim to answer two essential problems in GNN model design:

- **Q1: How does the P-T pattern affect multi-source UGDA?** i.e., between the P-T and T-P patterns, which one yields better performance and theoretical guarantees?
- **Q2: Given a specific pattern, what design principles of a concise P-T pipeline should follow for effective cross-domain generalization?** i.e., should the source branch involve more P-layers to preserve discriminative information, or should the target branch involve more P-layers to reduce domain discrepancy?

Empirical Study. We first conduct an empirical analysis to answer these two questions, as shown in Figure 3. For three commonly used research citation datasets, we transfer knowledge from source graphs Citationv1 (C) and DBLPv7 (D) to the target graph ACMv9 (A). We implement five operations, namely (1) P-T, (2) PPPT, (3) PPPPT, (4) T-P, and (5) TTTP, respectively, in the source branch and the target branch. For instance, PPPPT-T means stacking five P-layers first and then one T-layer. As shown in Figure 3, we could answer the above questions as:

- **A1: The P-T pattern is superior to that of the T-P pattern, whether in the source branch or the target branch.** For instance, when the source and the target branch both are with the PPP-T pattern and the TTT-P pattern, the node classification accuracies are 75.62% vs. 69.43%, which shows 8.92% improvement, demonstrating the P-T pattern yields better performance.

- **A2: Given the P-T pattern, the design principle that the target branch involves more P-layers and the source branch involves fewer P-layers can effectively achieve cross-domain generalization.** For instance, we could observe that the PPP-T can outperform the PPPPT-T in the source branch, and the PPPPT-T can outperform the PPP-T in the target branch, which confirms that for the P-T pattern, adopting the design principle that the target branch involves more P-layers can effectively achieve cross-domain generalization.

Theoretical Analysis. We aim to answer these two questions from two views: (1) We prove why the P-T pattern is better, through it yields a tighter upper bound on the cross-domain discrepancy optimization (PROPOSITION 4.1); and (2) We demonstrate why the target branch requires more propagation layers by analyzing the trade-off between domain discrepancy reduction and class discriminability under increasing propagation depth (PROPOSITION 4.2).

PROPOSITION 4.1. *For the P-T pattern (performing P first and then T), we calculate the Wasserstein distance between the propagated features as:*

$$\text{Dist}_{\text{wa}}(\mathbf{Z}_s^{\text{PT}}, \mathbf{Z}_t^{\text{PT}}) \leq L \left[\text{Dist}_{\text{wa}}(\mathbf{H}_s^{-1}, \mathbf{H}_t^{-1}) + C \|\hat{\mathbf{A}}_s - \hat{\mathbf{A}}_t\|_2 \|\mathbf{H}_t^{-1}\|_2 \right]. \quad (2)$$

For the T-P pattern (performing T first and then P), the Wasserstein distance satisfies

$$\text{Dist}_{\text{wa}}(\mathbf{Z}_s^{\text{TP}}, \mathbf{Z}_t^{\text{TP}}) \leq L \text{Dist}_{\text{wa}}(\mathbf{H}_s^{-1}, \mathbf{H}_t^{-1}) + C \|\hat{\mathbf{A}}_s - \hat{\mathbf{A}}_t\|_2 \mathbb{E}[\|\delta(\mathbf{H}^{-1}\mathbf{W}')\|_2]. \quad (3)$$

Since activation functions typically preserve or amplify norms, i.e.,

$$\mathbb{E}[\|\delta(\mathbf{H}^{-1}\mathbf{W}')\|_2] \geq \|\mathbf{H}_t^{-1}\|_2. \quad (4)$$

Hence, the upper bound for $\text{Dist}_{\text{wa}}(\mathbf{Z}_s^{\text{PT}}, \mathbf{Z}_t^{\text{PT}})$ is tighter than that for $\text{Dist}_{\text{wa}}(\mathbf{Z}_s^{\text{TP}}, \mathbf{Z}_t^{\text{TP}})$, demonstrating the superiority of the P-T pattern in reducing cross-domain distribution shift.

PROOF. For the P-T pattern yielding $\mathbf{Z}^{\text{PT}} = \delta(\hat{\mathbf{A}}\mathbf{H}^{-1}\mathbf{W}')$, note that T, with δ being a Lipschitz activation and $\mathbf{W}'$ a linear map with $\|\mathbf{W}'\|_2 = L$, satisfies the Lipschitz condition $\text{Lip}(T) \leq L$, where L is the Lipschitz constant of the spectral norm of $\mathbf{W}'$. By the data processing inequality for Wasserstein distance, we have:

$$\text{Dist}_{\text{wa}}(\mathbf{Z}_s^{\text{PT}}, \mathbf{Z}_t^{\text{PT}}) \leq L \text{Dist}_{\text{wa}}(\hat{\mathbf{A}}_s \mathbf{H}_s^{-1}, \hat{\mathbf{A}}_t \mathbf{H}_t^{-1}). \quad (5)$$

Using the triangle inequality to decompose the difference in normalized adjacency matrices $\hat{\mathbf{A}}_s$ and $\hat{\mathbf{A}}_t$, and leveraging the fact that $\|\hat{\mathbf{A}}\|_2 \leq 1$, we derive

$$\text{Dist}_{\text{wa}}(\hat{\mathbf{A}}_s \mathbf{H}_s^{-1}, \hat{\mathbf{A}}_t \mathbf{H}_t^{-1}) \leq \text{Dist}_{\text{wa}}(\mathbf{H}_s^{-1}, \mathbf{H}_t^{-1}) + C \|\hat{\mathbf{A}}_s - \hat{\mathbf{A}}_t\|_2 \|\mathbf{H}_t^{-1}\|_2. \quad (6)$$

For the T-P pattern yielding $\mathbf{Z}^{\text{TP}} = \hat{\mathbf{A}}\delta(\mathbf{H}^{-1}\mathbf{W}')$, we similarly obtain the following, with the Lipschitz property of δ :

$$\begin{aligned} \text{Dist}_{\text{wa}}(\mathbf{Z}_s^{\text{TP}}, \mathbf{Z}_t^{\text{TP}}) &\leq \text{Dist}_{\text{wa}}(\hat{\mathbf{A}}_s \delta(\mathbf{H}_s^{-1}\mathbf{W}'), \hat{\mathbf{A}}_t \delta(\mathbf{H}_t^{-1}\mathbf{W}')), \\ \text{Dist}_{\text{wa}}(\delta(\mathbf{H}_s^{-1}\mathbf{W}'), \delta(\mathbf{H}_t^{-1}\mathbf{W}')) &\leq L \text{Dist}_{\text{wa}}(\mathbf{H}_s^{-1}, \mathbf{H}_t^{-1}), \end{aligned} \quad (7)$$

and decomposing the adjacency matrix difference introduces a term involving $\mathbb{E}[\|\delta(\mathbf{H}^{-1}\mathbf{W}')\|_2]$, leading to Eq. (3). Finally, considering the activation function's property in Eq. (4), we could prove the operator-difference term in the T-P upper bound is larger than that in the P-T upper bound. $\square$

PROPOSITION 4.2. *Let $f = f_{\text{feat}} \circ f_{\text{cls}}$ denotes a hypothesis, where $f_{\text{feat}} : \mathcal{G} \rightarrow \mathbb{R}^d$ is a feature extractor and $f_{\text{cls}} : \mathbb{R}^d \rightarrow \{1, \dots, C\}$ is a classifier. For M source domains with distributions $\mathbb{P}_{s^1}, \dots, \mathbb{P}_{s^M}$ and empirical risks $\epsilon_{s^m}(f)$, there exists the target risk with weights $\alpha_m \geq 0$ and $\sum_{m=1}^M \alpha_m = 1$:*

$$\epsilon_t(f) \leq \sum_{m=1}^M \alpha_m \epsilon_{s^m}(f) + 2L_f L_c \sum_{m=1}^M \alpha_m \mathcal{D}(\mathbb{P}_{s^m}, \mathbb{P}_t) + \Omega. \quad (8)$$

Hence, using fewer P-layers in source domains and more P-layers in the target domain can reduce the target risk.

PROOF. Let $\epsilon_t(f) = \mathbb{E}_{(\mathcal{G}_t, \mathbf{y}_t) \sim \mathbb{P}_t}[\ell(f(\mathcal{G}_t), \mathbf{y}_t)]$ be the target risk, where ℓ is a loss function bounded by PROPOSITION 4.1. Assume f_{feat} is L_f -Lipschitz continuous and f_{cls} is L_c -Lipschitz continuous, empirical risks are $\epsilon_{s^m}(f) = \frac{1}{|\mathcal{V}_{s^m}|} \sum_{v \in \mathcal{V}_{s^m}} \ell(f(\mathcal{G}_{s^m}[v]), \mathbf{y}_{s^m}[v])$. This leads to inequality (8), where $\mathcal{D}(\mathbb{P}_{s^m}, \mathbb{P}_t)$ is a distribution divergence metric between the m -th source and target domains, and Ω is a constant accounting for irreducible label noise.

Using k_s and k_t P-layers on source graphs and the target graph, respectively, the Lipschitz constant L_f of f_{feat} is bounded by:

$$(L_f \propto \max(\|(\hat{\mathbf{A}}_s)^{k_s}\|_{\text{op}}, \|(\hat{\mathbf{A}}_t)^{k_t}\|_{\text{op}}), \quad (9)$$

where $\|\cdot\|_{\text{op}}$ denotes the operator norm, and $\|\hat{\mathbf{A}}^k\|_{\text{op}} \leq \lambda_{\max}(\hat{\mathbf{A}})^k \leq 1$ due to the property that $\lambda_{\max}(\hat{\mathbf{A}}) \leq 1$. L_f is influenced by the number of P-layers in both source and target graphs. Thus, decreasing k_s can prevent excessive amplification of the noise specific to the source domain, while increasing k_t will make $\lambda_{\max}(\hat{\mathbf{A}}_t)^{k_t}$ smaller, lowering L_f , further reducing the target risk. $\square$

4.2 Dual-Branch GNN with Concise Propagation–Transformation Pipeline

In our dual-branch GNN architecture, we stack fewer propagation layers on multiple source graphs and more propagation layers on the target graph, and such an asymmetric design allows the model to become more stable against graph distribution shifts, enhancing the generalization ability. Specifically, for each source graph $\mathcal{G}_s^m$ with a node feature matrix $\mathbf{X}_s^m \in \mathbb{R}^{|\mathcal{V}_s^m| \times d}$, the corresponding normalized adjacency matrices $\hat{\mathbf{A}}_s^m$ can be constructed as

$$\hat{\mathbf{A}}_s^m = \tilde{\mathbf{D}}_s^{m-1/2} \cdot \tilde{\mathbf{A}}_s^m \cdot \tilde{\mathbf{D}}_s^{m-1/2}, \quad (10)$$

where $\tilde{\mathbf{A}}_s^m = \mathbf{A}_s^m + \mathbf{I}$ with $\mathbf{I}$ being the identity matrix to include self-loops and $\tilde{\mathbf{D}}_s^m$ is the corresponding degree matrix. Similarly, for the target graph $\mathcal{G}_t$, we have the normalized adjacency matrix $\hat{\mathbf{A}}_t$ and feature matrix $\mathbf{X}_t \in \mathbb{R}^{|\mathcal{V}_t| \times d}$. Therefore, the source branch with k_s P-layers and one T-layer followed can be denoted as:

$$\begin{aligned} \mathbf{H}_s^{(l,m)} &= \text{TP}^{(l)}(\mathbf{X}_s^m) = \text{T} \left[\underbrace{\text{P}(\text{P}(\dots \text{P}(\mathbf{X}_s^m)) \dots)}_{k_s \times \text{P}} \right] \\ &= \sigma[(\hat{\mathbf{A}}_s^m)^{k_s} \cdot \mathbf{X}_s^m \cdot \mathbf{W}_s^m], \end{aligned} \quad (11)$$

where $l \in \{1, 2, \dots, k_s\}$ denotes the propagation layers, $\sigma(\cdot)$ is the non-linear activation function, and $\mathbf{H}_s^{(l,m)}$ is the node representation matrix from the source branch. Note that $\mathbf{W}_s^m$ is a learnable weight matrix for the m -th source domain. Then, we use k_t P-layers and one T-layer on the target branch, and importantly, we set $k_t > k_s$ to avoid over-fitting source-specific topological structures and over-smoothing of source-specific node features; Meanwhile, the complex topological patterns critical for target-specific tasks can be captured to enhance the model domain adaptation ability on the target graph. Hence, we have:

$$\begin{aligned} \mathbf{H}_t^{(l)} &= \text{TP}^{(l)}(\mathbf{X}_t) = \text{T} \left[\underbrace{\text{P}(\text{P}(\dots \text{P}(\mathbf{X}_t)) \dots)}_{k_t \times \text{P}} \right] \\ &= \sigma[(\hat{\mathbf{A}}_t)^{k_t} \cdot \mathbf{X}_t \cdot \mathbf{W}_t], \end{aligned} \quad (12)$$

where $\hat{\mathbf{A}}_t$ denotes the normalized target adjacency matrix, and $\mathbf{W}_t$ denotes the learnable weight matrix of the target graph.

4.3 Domain Alignment for Node Classification

To efficiently explore useful knowledge between graphs, we design a comprehensive loss function that integrates three different loss functions to ensure effective node classification and domain alignment. The overall loss function is formulated as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}} + \lambda \mathcal{L}_{\text{align}} + \delta \mathcal{L}_{\text{ps}}, \quad (13)$$

where λ, δ are the balance parameters, and the $\mathcal{L}_{\text{ce}}$, $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{ps}}$ denote the source classifier loss, the alignment loss and the pseudo-label loss, respectively. The details are introduced as follows.

4.3.1 Source Classifier Loss. We deliver $\mathbf{H}_s^m$ into the classifier $f_{(s,\text{cls})}^m(\cdot)$ for the m -th source graph, and the class predictions are denoted as

$$\hat{\mathbf{Y}}_s^m = f_{(s,\text{cls})}^m(\mathbf{H}_s^m; \mathbf{W}_{(s,\text{cls})}^m), \quad (14)$$

where $\mathbf{W}_{(s,\text{cls})}^m \in \mathbb{R}^{d \times C}$ represents the learnable weight matrix of the classification layer with d as the input feature dimension and C as the number of classes. For M labeled graphs $\mathcal{G}_s = (\mathcal{V}_s^m, \mathcal{E}_s^m, \mathbf{Y}_s^m)_{m=1}^M$ from different source domains, we use the cross-entropy loss to minimize the source classifier loss, defined as

$$\mathcal{L}_{\text{ce}} = -\frac{1}{M} \sum_{m=1}^M \frac{1}{|\mathcal{V}_s^m|} \sum_{v \in \mathcal{V}_s^m} \sum_{c=1}^C \mathbf{Y}_s^m(v, c) \cdot \log(\hat{\mathbf{Y}}_s^m(v, c)), \quad (15)$$

where $\mathbf{Y}_s^m(v, c)$ denotes the ground-truth label of node v in the m -th source graph, and $\hat{\mathbf{Y}}_s^m(v, c)$ denotes the classification prediction of node v in the m -th source graph.

4.3.2 Alignment Loss Weighted by Graph Wasserstein Distance. To quantify and minimize structural and feature distribution gaps between source and target graphs, we measure the alignment loss between the source and target graphs using the MMD technique [11], weighted by the graph Wasserstein distance [33]. For each source graph $\mathcal{G}_s^m$, we compute the graph Wasserstein distance between each source graph and the target graph as:

$$\gamma_{\text{wa}}^m = \text{Dist}_{\text{wa}}(\mathcal{G}_s^m, \mathcal{G}_t), \quad (16)$$

which quantifies the structural and feature-level divergence between the source and target graphs. By incorporating this distance

into the learning process, we explicitly encode the inherent source-target domain differences, thereby enabling the model to dynamically adjust the contribution of each source graph and learn representations that better conform to the structure and features of the target graph. Hence, the alignment loss is defined as:

$$\mathcal{L}_{\text{align}} = \sum_{m=1}^M \gamma_{\text{wa}}^m \cdot \text{MMD}(\mathbf{H}_s^m, \mathbf{H}_t), \quad (17)$$

where $\text{MMD}(\cdot)$ denotes the maximum mean discrepancy to measure the distribution shift in the latent feature space.

4.3.3 Pseudo-label Loss. To optimize the model's parameters, we generate target pseudo-labels using class predictions denoted as

$$\mathbf{P}_t = f_{(t, \text{cls})}(\mathbf{H}_t; \mathbf{W}_{(t, \text{cls})}), \quad (18)$$

where $\mathbf{P}_t = \{\mathbf{p}_{(i,t)}\}_{i=1}^{|\mathcal{V}_t|}$, and $f_{(t, \text{cls})}(\cdot)$ is the target graph classifier and $\mathbf{W}_{(t, \text{cls})}$ represents the learnable weight matrix. Then, we maintain two memory banks for the target graph to stabilize pseudo-labels generation. The first is a representation bank $\mathcal{B} = [\tilde{\mathbf{b}}_1, \dots, \tilde{\mathbf{b}}_{|\mathcal{V}_t|}]$, where $\tilde{\mathbf{b}}_i \in \mathbb{R}^d$ denotes the node representation of the i -th target node initialized by $\tilde{\mathbf{b}}_i^{k=0} = \mathbf{h}_i \in \mathbf{H}_t$. The second is a prediction bank $\mathcal{P} = [\tilde{\mathbf{p}}_1, \dots, \tilde{\mathbf{p}}_{|\mathcal{V}_t|}]$ denotes the predicted class distribution of the i -th target node initialized by $\tilde{\mathbf{p}}_i^{k=0} = \mathbf{p}_{(i,t)}$. These banks are updated K rounds with $\{k = 1, 2, \dots, K\}$ via a momentum mechanism, and for k -th update, we have:

$$\tilde{\mathbf{b}}_i^{k+1} = (1 - \alpha)\tilde{\mathbf{b}}_i^{k-1} + \alpha\tilde{\mathbf{b}}_i^k, \quad \tilde{\mathbf{p}}_i^{k+1} = (1 - \alpha)\tilde{\mathbf{p}}_i^{k-1} + \alpha\tilde{\mathbf{p}}_i^k, \quad (19)$$

where $\tilde{\mathbf{b}}_i^k$ and $\tilde{\mathbf{p}}_i^k$ are current representations and predictions of the i -th target node for each training, $\tilde{\mathbf{b}}_i^{k-1}$ and $\tilde{\mathbf{p}}_i^{k-1}$ are from last update, $\tilde{\mathbf{b}}_i^{k+1}$ and $\tilde{\mathbf{p}}_i^{k+1}$ are the results of update, and α is a smoothing parameter. For each node i , we retrieve r nearest neighbors from $\mathcal{B}$ based on cosine similarity, defined as:

$$\text{cos_sim}(\tilde{\mathbf{b}}_i, \tilde{\mathbf{b}}_j) = \frac{\tilde{\mathbf{b}}_i^\top \tilde{\mathbf{b}}_j}{\|\tilde{\mathbf{b}}_i\|_2 \|\tilde{\mathbf{b}}_j\|_2}, \quad Q(i) = \{j \mid \tilde{\mathbf{b}}_j \in \mathcal{N}_r(\tilde{\mathbf{b}}_i)\}, \quad (20)$$

where $Q(i)$ is the set of indices of r nearest neighbors for the i -th node. The pseudo-label distribution for the i -th node is formed by aggregating the predicted distributions of these neighbors from $\mathcal{P}$:

$$\hat{\mathbf{y}}_i = \mathbb{H}[\arg \max_c \left(\frac{1}{|Q(i)|} \sum_{j \in Q(i)} \tilde{\mathbf{p}}_j \right)], \quad (21)$$

where $\mathbb{H}[\cdot]$ denotes the one-hot transformation. We optimize the model by minimizing the cross-entropy loss between the generated pseudo-labels $\hat{\mathbf{y}}_i$ and predicted distributions $\mathbf{p}_{(i,t)}$:

$$\mathcal{L}_{\text{ps}} = -\frac{1}{|\mathcal{V}_t|} \sum_{i=1}^{|\mathcal{V}_t|} \sum_{c=1}^C \hat{\mathbf{y}}_{i,c} \log(\mathbf{p}_{i,t}^c), \quad (22)$$

where $\hat{\mathbf{y}}_{i,c}$ is the c -th element of the one-hot pseudo-label $\hat{\mathbf{y}}_i$, and $\mathbf{p}_{i,t}^c$ is the c -th element of predicted distribution $\mathbf{p}_{(i,t)}$.

4.4 Time Complexity Analysis

Consider a graph with $|\mathcal{V}|$ nodes and $|\mathcal{E}|$ edges, where the node representation dimension is set to d . When k P-layers are applied, the time complexity of message passing via a sparse adjacency matrix is $O(M \cdot k \cdot |\mathcal{E}| \cdot d)$ with M source domains. The time complexity

of cross-entropy loss for the source domains is $O(M \cdot |\mathcal{V}| \cdot C)$, where C denotes the number of categories. For the alignment loss calculation, the time complexity is $O(M \cdot |\mathcal{V}|^2 \cdot d)$. Additionally, generating pseudo-labels by finding nearest neighbors requires a time complexity of $O(d \cdot |\mathcal{N}_r| \log |\mathcal{N}_r|)$, where $|\mathcal{N}_r|$ refers to the total number of nodes involved in the nearest neighbors.

5 Experiments

5.1 Experimental Settings

Datasets. To evaluate the effectiveness of our proposed CPT-GDA, we conduct node classification tasks on three different types of graph datasets: (1) **CSBM** [6] consists of four graphs: C1, C2, C3, and C4, which are synthetic graphs generated by a 4-class contextual stochastic block model [51]; (2) **Research Citation Networks** [31] includes three research paper citation networks, i.e., DBLPv7 (D), ACMv9 (A), and Citationv1 (C), collected from distinct platforms and timeframes. These datasets represent academic citation networks, where nodes are papers and edges are citation relationships; (3) **Twitch** [26] is a social media graph dataset, which aggregates six social networks from different regions: Germany (DE), the United Kingdom (EN), Spain (ES), France (F), Portugal (P), and Russia (R). The statistical summary of the datasets is shown in the Appendix.

Baseline Methods. The proposed method is compared with representative and state-of-the-art adaptation approaches. We compare our proposed method with two groups of baselines: (i) *Single-source adaptation methods*: UDAGCN [40], GRADE [39], SpecReg [47], SHOT [19], BNM [5], ATDOC [20], NRC [46], JMDS [16], GTrans [14], GraphCTA [52] and TPDS [32]. We merge all samples from different source domains into a unified source domain to adapt to multi-source domain scenarios. (ii) *Multi-source adaptation methods*: MDAN [53], DECISION [1], CAiDA [7], MSFDA [27] and GraphATA [51]. We make necessary adaptations of CV models to the graph structural data. Each experiment is repeated five times, and we report the average accuracy and standard deviation.

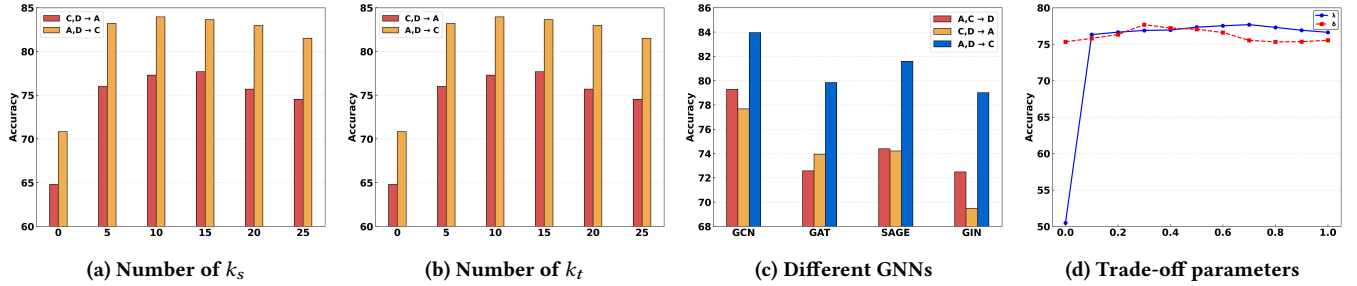
Implementation. Following works [28, 40], we use 80% of the labeled source nodes to provide supervision, while the remaining 20% labeled source nodes form the validation set. The evaluation is conducted on the target nodes. For comparisons with baselines, we adopt the publicly available code provided by the authors and retain the same GNN backbone with identical layers. The node representation dimension is set as 128 for all the methods. Our proposed method is implemented using Pytorch [25]. The optimal learning rate and weight decay are searched in the set of $\{1e-1, 1e-2, 1e-3, 1e-4, 5e-4\}$. The smoothing parameter α for the memory banks is set to 0.9 by default, and the trade-off parameters λ and δ are determined via a search within the range $[0, 1]$.

5.2 Experimental Results

We present the results of node classification in Table 1. Firstly, the results of most multi-source methods are superior to those of single-source methods. The advantage of multi-source methods stems from their effective utilization of complementary knowledge across multiple source domains. Secondly, when further compared with other multi-source methods, our method can still outperform the best model GraphATA[51] by an average of 3%. It indicates that the concise propagation-transformation pipeline proves to be a

Table 1: Performance comparison of node classification accuracy (ACC $\pm$ std%).

Types	Methods	CSBM		Citation		Twitch	
		C1,C2,C3 $\rightarrow$ C4	A,C $\rightarrow$ D	C,D $\rightarrow$ A	A,D $\rightarrow$ C	R,P,F,ES $\rightarrow$ DE	R,P,F,ES $\rightarrow$ EN
Single-Source	UDAGCN [40]	72.18 $\pm$ 0.12	70.54 $\pm$ 0.04	61.67 $\pm$ 0.01	69.72 $\pm$ 0.03	43.34 $\pm$ 0.30	48.88 $\pm$ 0.22
	GRADE [39]	78.12 $\pm$ 0.21	74.37 $\pm$ 0.49	70.92 $\pm$ 0.04	79.48 $\pm$ 0.05	53.52 $\pm$ 0.07	46.80 $\pm$ 0.04
	SpecReg [47]	79.14 $\pm$ 0.38	76.42 $\pm$ 0.66	72.69 $\pm$ 0.51	78.67 $\pm$ 2.46	42.85 $\pm$ 2.72	47.21 $\pm$ 1.34
	SHOT [19]	83.82 $\pm$ 0.25	75.34 $\pm$ 0.13	68.71 $\pm$ 0.41	77.92 $\pm$ 0.05	64.01 $\pm$ 0.09	57.97 $\pm$ 0.07
	BNM [5]	81.79 $\pm$ 0.47	75.17 $\pm$ 0.16	68.81 $\pm$ 0.12	77.38 $\pm$ 0.11	64.11 $\pm$ 0.23	57.79 $\pm$ 0.20
	ATDOC [20]	64.72 $\pm$ 0.33	72.67 $\pm$ 0.63	65.52 $\pm$ 1.16	76.12 $\pm$ 0.29	57.92 $\pm$ 0.97	50.94 $\pm$ 2.80
	NRC [46]	84.42 $\pm$ 0.16	73.13 $\pm$ 0.08	69.52 $\pm$ 0.25	77.51 $\pm$ 0.22	63.76 $\pm$ 0.09	57.66 $\pm$ 0.05
	JMDS [16]	63.74 $\pm$ 0.39	72.21 $\pm$ 0.37	65.26 $\pm$ 0.18	74.34 $\pm$ 0.29	64.00 $\pm$ 0.05	46.78 $\pm$ 0.17
	GTrans [14]	64.81 $\pm$ 0.89	73.67 $\pm$ 0.65	65.59 $\pm$ 2.06	78.74 $\pm$ 0.43	62.54 $\pm$ 0.02	57.29 $\pm$ 0.07
	GraphCTA [52]	83.21 $\pm$ 0.43	75.53 $\pm$ 0.30	70.91 $\pm$ 0.63	79.59 $\pm$ 0.22	62.70 $\pm$ 0.24	56.28 $\pm$ 0.16
Multi-Source	TPDS [32]	84.03 $\pm$ 0.23	71.63 $\pm$ 0.12	68.45 $\pm$ 0.23	75.87 $\pm$ 0.21	63.14 $\pm$ 0.10	57.83 $\pm$ 0.14
	MDAN [53]	76.49 $\pm$ 0.39	70.38 $\pm$ 0.17	65.31 $\pm$ 0.09	73.92 $\pm$ 0.13	61.78 $\pm$ 0.33	49.52 $\pm$ 0.35
	DECISION [1]	88.02 $\pm$ 0.28	76.02 $\pm$ 0.67	70.40 $\pm$ 0.31	79.39 $\pm$ 0.35	63.92 $\pm$ 0.26	57.92 $\pm$ 0.12
	CAiDa [7]	87.96 $\pm$ 0.94	75.75 $\pm$ 0.42	70.25 $\pm$ 0.54	79.70 $\pm$ 0.74	63.73 $\pm$ 0.49	57.73 $\pm$ 0.45
	MSFDA [27]	88.94 $\pm$ 0.62	76.03 $\pm$ 1.27	68.60 $\pm$ 2.94	78.63 $\pm$ 0.92	54.42 $\pm$ 5.50	55.77 $\pm$ 0.58
	GraphATA [51]	90.14 $\pm$ 0.36	78.45 $\pm$ 0.87	73.17 $\pm$ 0.52	82.33 $\pm$ 0.75	66.71 $\pm$ 0.39	59.56 $\pm$ 0.14
CPT-GDA (Ours)		91.23$\pm$0.20	79.28$\pm$0.43	77.68$\pm$0.25	83.95$\pm$0.34	67.45$\pm$0.23	60.43$\pm$0.09

**Figure 4: (a) shows the accuracy of the number of propagation layers k_s . (b) shows the accuracy of the number of propagation layers k_t . (c) shows the accuracy of different GNN architectures. (d) shows the sensitivity of trade-off parameters λ and δ .**

crucial design choice. Using fewer propagation layers for source domains avoids overfitting to the topologies of individual sources, while using more propagation layers for the target domain can capture the complex structural features required for precise node classification. This strategic balance allows our model to handle the varying graph topologies encountered in different domains with greater proficiency. Thirdly, our proposed CPT-GDA consistently outperforms all baselines with a significant advantage in various adaptation tasks. For instance, we achieve 13.65% average improvements in the scenario of C, D $\rightarrow$ A compared with all baselines. We attribute these expressive results to the new loss function strategy, which can effectively mine the useful knowledge between source and target graphs and reduce domain differences more effectively than traditional methods. In conclusion, our proposed CPT-GDA demonstrates impressive performance across different datasets and tasks, thus verifying its capability in multi-source graph domain adaptation tasks.

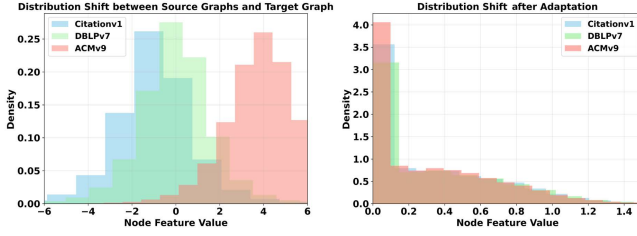
5.3 In-depth Experimental Analysis

Ablation Studies of Different Submodules. To comprehensively investigate the contribution of each constituent component within our proposed CPT-GDA model, a series of ablation experiments

is carried out on citation datasets. Table 2 presents the ultimate results obtained when incorporating each module, with no performance degradation observed. Specifically, we consider four ablation cases: (1) w/o P-T pipeline, which uses a simple GCN model in each branch, along with only $\mathcal{L}_{ce}$ as the classifier loss; (2) + P-T pipeline, which uses concise P-T pipeline; (3) + $\mathcal{L}_{ps}$, which adds to the pseudo-label loss; and (4) + $\mathcal{L}_{align}$, which adds to the alignment loss. The first line, $\mathcal{L}_{ce}$, only considers the source domain information, resulting in unsatisfactory performance, indicating that this model cannot be effectively applied to the target graph due to the existence of domain shifts. Next, in the second line, the average results increase by the most significant 4.01% through the addition of a concise P-T pipeline, which confirms that our dual-branch GNN can effectively learn node representations to reduce domain divergence. The third line involves updating the model parameters by augmenting the pseudo-label loss, resulting in an improvement of 1.11%. This serves to boost the model’s adaptability to diverse graph domains. The results in the fourth line improve by 0.77%, indicating that calculating the weights of source domains reduces domain alignment loss. In conclusion, our proposed CPT-GDA attains the best performance, thus validating the effectiveness of integrating all the modules mentioned above.

Table 2: Ablation study of each submodule in CPT-GDA.

Modules	A,C→D	C,D→A	A,D→C
w/o P-T + $\mathcal{L}_{ce}$	75.46±0.63	72.59±0.62	79.28±0.59
+ P-T	78.42±0.53	75.54±0.42	82.50±0.52
+ $\mathcal{L}_{ps}$	78.86±0.49	76.48±0.31	83.75±0.42
+ $\mathcal{L}_{align}$ (Ours full.)	79.28±0.43	77.68±0.25	83.95±0.34

**Figure 5: Distribution shift visualization.**

The Impact of Propagation Layers. To analyze the effects of the propagation layers k_s and k_t , we conduct experiments on the tasks of $C, D \rightarrow A$ and $A, D \rightarrow C$, as shown in Figure 4 (a) and (b). On the one hand, restricting k_s to a small range ($k_s < 5$) yields a substantial boost in adaptation efficiency, with the peak performance consistently observed when $k_s = 3$. A small k_s alleviates the overfitting of topological noise specific to the source domain. When k_s increases to more than 3, the propagation operation amplifies the special structural pattern unique to a single source domain. These source-specific features are not transferable to the target domain, leading to degraded generalization, which is consistent with our theoretical analysis. On the other hand, different values of k_t produce different effects. When the value range of k_t is 5 to 15, the performance of the model is optimal, because the concise P-T pipeline effectively aggregates the topological information of the target graph. When k_t exceeds 15, the performance of the model shows a significant downward trend. This might be due to the fact that excessive propagation increases the risk of propagating noisy structural information in the target graph, further degrading the quality of the learned features and weakening the model’s ability to align with the source domains.

The Effect of Different GNN Architectures. To explore the transferability of prevalent graph neural network architectures in graph transfer learning scenarios, we conduct comparative experiments on four widely used models: GCN [15], GAT [34], SAGE [12], and GIN [45]. The experimental results are presented in Figure 4 (c). Notably, the simplest GCN architecture achieves the best performance among all tested models, suggesting that overly complex propagation mechanisms may not always bring additional benefits to cross-domain graph representation transfer. On the contrary, the simplicity and efficiency of the message passing mechanism of GCN can better adapt to the distribution shifts between source and target graph domains.

The Sensitivity of Trade-off Parameters. Our proposed framework incorporates two key trade-off parameters, λ and δ , which respectively weight the alignment loss and the pseudo-label loss.

These parameters directly affect the optimization direction. λ controls the strength of domain adaptation to align source and target distributions, while δ adjusts the contribution of pseudo-label-guided learning to refine target domain classification. To assess their impact on model performance, we conduct experiments on the $C, D \rightarrow A$ task. Figure 4 (d) illustrates the accuracy of our model across different values of λ and δ . The results confirm that our model is robust to variations in trade-off parameters λ and δ within $[0.0, 1.0]$, while the optimal configuration ($\lambda = 0.7$, $\delta = 0.3$) strikes a balanced trade-off between domain alignment and pseudo-label supervision. This balance ensures the effective transfer of domain-invariant knowledge from the source domains to the target domain, while leveraging unlabeled target data to further refine classification performance.

Distribution Shift Visualization. To evaluate the adaptive effect on the target domain, we visualize the density distribution of node features, as shown in Figure 5. In the $C, D \rightarrow A$ scenario, the source domain graphs are Citationv1 (C) and DBLPv7 (D), and the target domain graph is ACMv9 (A). Before adaptation, there were obvious distribution differences between the source domain graphs and the target domain graph. After adaptation, there is a notable narrowing of the gap between source feature distributions and the target feature distribution. This indicates that the CPT-GDA we proposed successfully aligns the source features with the target features within this range, reducing the initial distribution shift. Furthermore, when we move to higher node feature values, the feature densities of C, D, and A show increasingly close consistency. Such alignment is crucial for domain adaptation tasks, as it ensures that models trained on source domains can generalize well to the target domain.

6 Conclusion

In this paper, we propose a novel multi-source unsupervised graph domain adaptation approach named **CPT-GDA**. We employ a concise P-T pipeline strategy to construct a dual-branch GNN architecture with different depths of propagation but the same fixed P-T patterns, which can better capture complementary knowledge from multiple sources in an efficient manner, as supported by our theoretical analysis. Meanwhile, we derive three optimization objectives, i.e., (1) *the classifier loss* using the source labeled data, which can learn discriminative representations; (2) *the alignment loss* weighted by the graph Wasserstein distance, which can align the structure and feature distribution to better utilize useful knowledge; and (3) *the pseudo-label loss* between generated target pseudo-labels and predictions distributions, which can refine target node representations. Extensive experiments on real-world datasets confirm the effectiveness of our proposed CPT-GDA. For future work, we will extend our model to more complex adaptive scenarios, such as open-set graph domain adaptation and source-free unsupervised graph domain adaptation.

Acknowledgments

This work is supported by the Fundamental Research Funds for the Central Universities (No. DUT25YG238).

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7 Appendix

Table 3: Dataset Statistics.

Datasets	#Nodes	#Edges	#Feat	#Class
CSBM	8,000~8,000	607,699~752,776	128	4
Twitch	1,912~9,498	31,299~153,138	3,170	2
Citation	5,484~9,360	8,117~15,556	6,775	5